Coverage, Entry, and Engineering Components

Three denominators, sparse first entry, and an exact accounting identity

PAPER TITLE

Coverage, Entry, and Engineering Components of Electricity Recovery in Japan's Municipal Waste-Incineration Fleet, FY2005-FY2024

PROFESSOR BRIEFING | 18-SLIDE ROUTE | DESCRIPTIVE + DIAGNOSTIC

Pann Phetra
Paper discussion briefing

The Paper Asks Three Different Questions

RQ1 · COVERAGE

How much of the fleet is covered?

Compare facility records, waste throughput, and processing design capacity.

RQ2 · ENTRY

Which non-generators first report installed capacity?

Use an at-risk history and a sparse-event estimator.

RQ3 · COMPONENTS

What produces gross electricity per tonne?

Separate installed sizing, annual capacity factor, and waste loading.

One fleet average cannot answer all three questions: the denominators, candidate populations, and outcomes differ.

Two FY2024 Count Statistics, Not One

PUBLISHED NATIONAL CONTEXT

415 / 991 = 41.9%

official electricity-generating facilities

The Ministry's published facility count and denominator.

ANALYTICAL PANEL

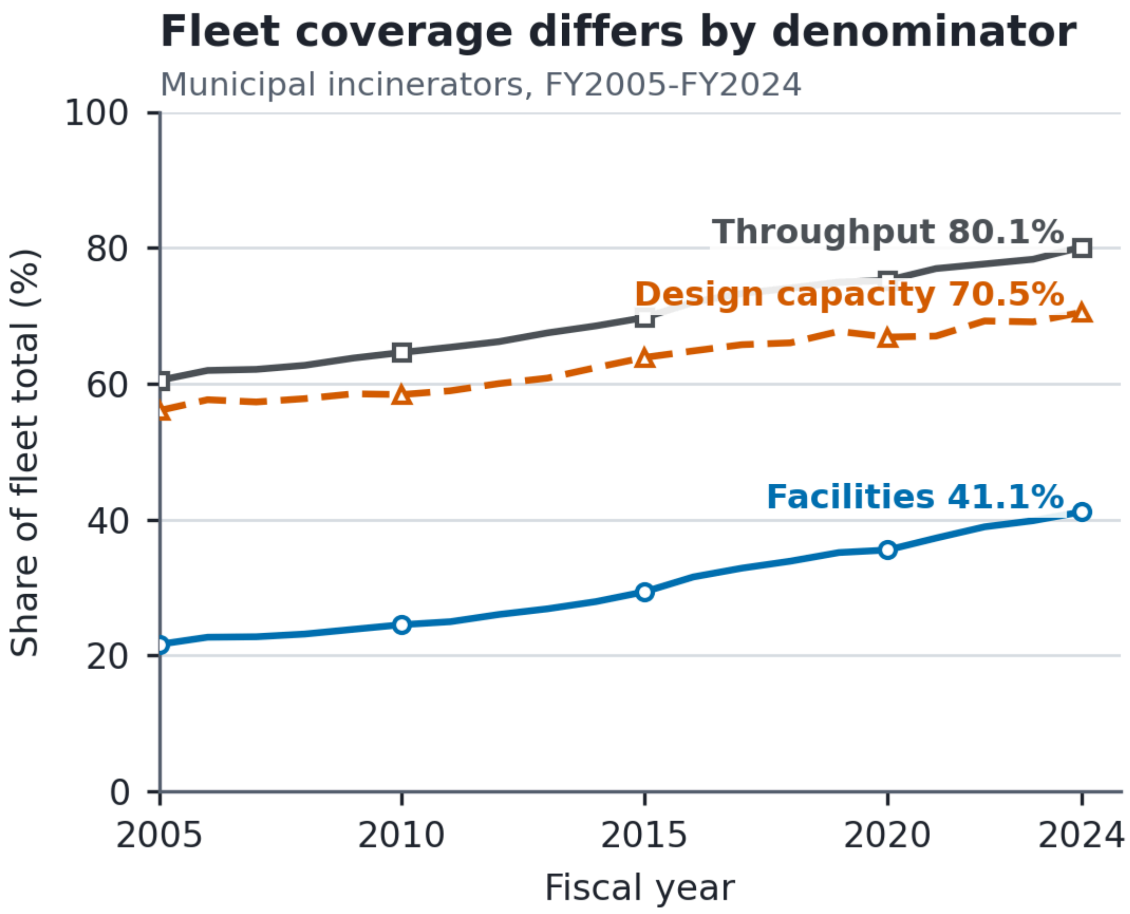
417 / 1,014 = 41.1%

retained records with installed capacity

The paper's record definition and reconstructed denominator.

The 41.9% official statistic is context; it is not substituted for the paper's 41.1% analytical measure.

Facility Counts Understate Waste Coverage



FACILITY PARTICIPATION

41.1%

Retained FY2024 records with installed capacity.

THROUGHPUT COVERAGE

80.1%

Recorded tonnes handled by positive-output facilities.

DESIGN-CAPACITY COVERAGE

70.5%

Processing capacity at installed-capacity facilities.

Audited Identity Comes Before Transitions

PARSED SOURCE

23,599

raw rows

Twenty annual workbooks, FY2005-FY2024.

→

DEDUPLICATED

23,593

retained records

Six exact duplicate source records collapsed.

→

LONGITUDINAL IDENTITY

1,690

stable administrative lineages

Deterministically linked record histories.

→

CONFIGURATION HISTORY

1,767

asset episodes

New episode when reported asset evidence resets.

A **stable administrative lineage** is an inferred administrative history, not proof of unchanged physical assets. All 16 accepted uncertain links are exposed and tested by whole-lineage exclusion.

Analytical Frames Match the Estimands

Frame	Rows	Lineages	Events	Purpose
Descriptive entry risk set	16,519	1,223	55	First reported installed-capacity events
Broad exact-year Firth	15,154	1,137	35	Adjacent-year, complete-covariate entry model
Prior-operation Firth	13,072	1,019	33	Exact frame plus positive prior throughput
Same-episode / identity-certain	15,095 / 15,107	1,135 / 1,130	24 / 35	Continuity / linkage sensitivities
Engineering components	6,511	493	-	Positive-throughput, positive-output generator-years within bounds

Entry means the **first report of positive installed electrical capacity**; it is not a verified retrofit or commissioning date.

Why Firth for First Entry?

SPARSE-EVENT ESTIMATOR

Firth = bias-reduced logistic regression using a Jeffreys-prior penalty

It reduces first-order small-sample bias and keeps estimates finite under separation.

EXACT-YEAR FRAME

Only adjacent records still at risk.

Prior age and processing capacity describe the lineage before entry.

PRIOR-OPERATION FRAME

Require positive prior-year throughput.

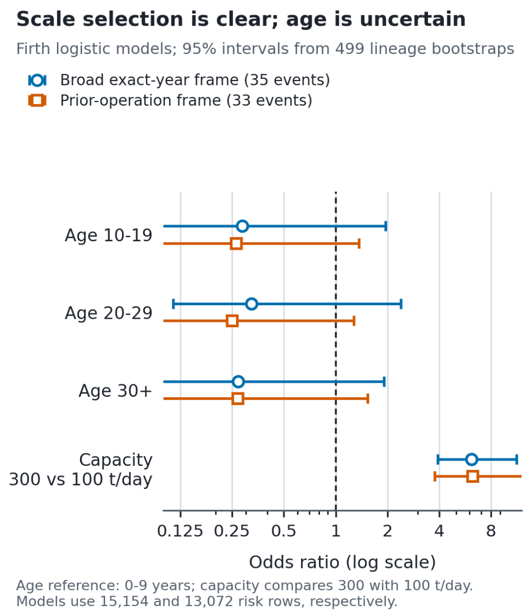
This is a nested sensitivity, not an independent comparison group.

INTERPRETATION

Association, not intervention effect.

All four models complete 499 whole-lineage bootstrap replications.

Scale Selection Is Clear; Age Is Uncertain



SCALE • 300 VS 100 T/DAY

OR 6.13 / 6.25

Strong positive association in both entry frames.

JOINT AGE EVIDENCE

.380 / .186

Broad / prior-operation joint age p-values. Same episode: .051; identity certain: .357.

What the Entry Result Supports

SUPPORTED

Entry is rare and strongly scale-selective.

A 300-versus-100 t/day contrast has about six times the conditional odds in both frames.

NOT SUPPORTED

No defensible age-only rule.

Broad and identity-certain tests are null; the 24-event same-episode result is borderline.

OBSERVED PATHWAYS · 55 EVENTS

35 continuity · 11 rebuild-like · 9 forward-dated

Administrative classifications describe evidence patterns, not verified projects.

Processing scale is a screening marker for unobserved technical, financial, and municipal conditions, not the causal effect of enlarging a plant.

Gross MWh/t Is an Accounting Identity

EXACT FACILITY-YEAR IDENTITY

gross MWh/t = 0.024 × design intensity × capacity factor ÷ processing utilization

Gross MWh/t = annual gross electricity generated divided by annual waste throughput.

Design intensity = installed electrical kW per t/day of waste-processing design capacity.

Capacity factor = annual gross output relative to installed kW operating for all 8,760 hours.

Processing utilization is annual throughput divided by 365 times processing design capacity. The identity is exact, but it is not a causal decomposition or an independent efficiency score.

Engineering Frame Separates Design From Operation

PRIMARY COMPONENT FRAME

6,511

generator-years

Positive capacity, throughput, and output within predeclared engineering bounds.

REPEATED HISTORIES

493

stable lineages

Uncertainty is clustered by audited stable lineage.

TWO COMPONENT MODELS

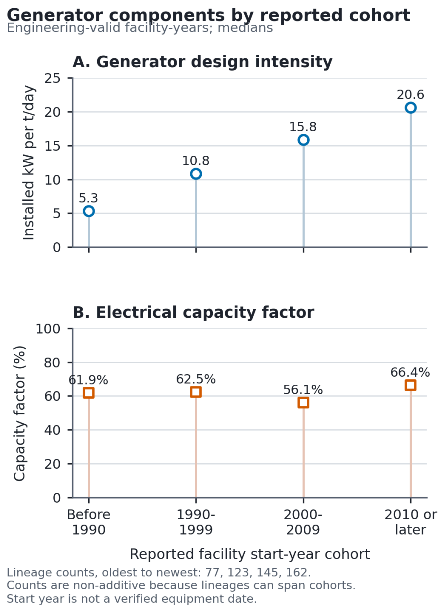
D + F

installed sizing + annual use

Waste loading enters through processing utilization.

Reported start year is an administrative **design-vintage marker**, not a verified boiler or generator installation date.

The Cohort Hierarchy Is Mainly Generator Sizing



DESIGN INTENSITY

5.3 → 20.6

Median installed kW per t/day, oldest to newest reported cohort.

CAPACITY FACTOR

No monotonic gradient

Annual use of installed kW does not reproduce the pattern; the hierarchy is principally installed sizing.

The Age Association Is Specification-Dependent

HEATING-CONTROLLED LEGACY SPECIFICATION

age = -0.0349

$p<.001$ • R^2=.4737

Gross MWh/t appears lower with reported age when installed sizing is omitted.

ADD GENERATOR DESIGN INTENSITY

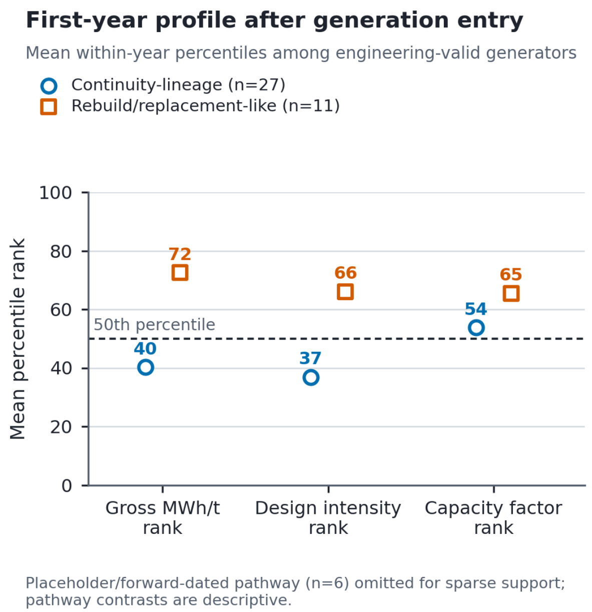
age = -0.0020

p=.2977 • R^2=.8131

Design intensity enters at +0.7532 ($p<.001$).

The diagnostic uses **5,806** engineering-valid rows with plausible heating value. Scale and utilization also become non-significant after sizing is added.

Observed Pathways Start at Different Ranks



CONTINUITY · N=27

40 / 37 / 54

Gross MWh/t / design intensity / capacity factor ranks.

REBUILD-LIKE · N=11

72 / 66 / 65

Same first-complete-year ranks. Descriptive selected contrast; six forward-dated observations are omitted.

Stress Tests Preserve the Component Interpretation

Stress test	Current evidence	Interpretation
Two ten-year windows	Scale coefficient in the design-intensity model: 0.474 / 0.577	The sizing relationship appears in both periods.
Engineering bounds	Conservative / broad scale coefficients: 0.520 / 0.536	The result is not tied to one plausibility range.
Lineage-equal weighting	Design-vintage hierarchy remains	Longer observed histories do not create the main pattern.
Identity-certain lineages	6,450 rows; scale coefficient 0.533	Accepted uncertain links do not drive the component pattern.
Within-episode operating models	Utilization-capacity-factor association remains positive	Used only for components with meaningful within-asset change.

Robustness strengthens the descriptive component result; it does not make throughput, maintenance, sizing, or output exogenous.

Evidence Boundaries Are Part of the Result

IDENTITY

Lineages are inferred.

No physical-site registry verifies ownership, equipment, replacement, or closure histories.

ENTRY

35 broad; 24 same episode.

Firth and bootstrap intervals cannot replace missing project, finance, or contract data.

ENGINEERING

Gross output is not net benefit.

No complete net export, useful heat, cost, outage, or lifecycle-emissions measure.

All estimates are descriptive or diagnostic associations. The paper does not identify retrofit effects, pathway effects, or an optimal investment policy.

Contribution: Measurement Before Mechanism

COVERAGE

Report counts and volumes together.

A 41.1% record share coexists with 80.1% throughput coverage.

ENTRY

Use scale for screening, not an age-only rule.

Scale is strongly associated with entry; joint age evidence is uncertain.

GENERATORS

Decompose before benchmarking.

Gross MWh/t should be read through sizing, capacity factor, and waste loading.

> The next evidence should link procurement, construction, generator history, net export, heat use, outages, waste composition, and finance to specific projects.

Discussion Questions

The paper separates fleet coverage, first entry, and generator components before interpreting hierarchy.

Is the three-estimand architecture convincing? Are the Firth entry design and engineering identity explained clearly enough? Which missing project-level evidence should come next?